University of Manitoba

ECE 4860-T16: Generative AI **Winter 2026**

Assignment 2: VAEs

February 2026

The purpose of this assignment is to give you hands-on experience designing and training variational auto-encoders.

Due Data: The assignment is due March 1st, 2026 by 11:59 pm (electronic submission through UMLearn).

Assignments should be performed individually, but feel free to interact with your classmates. Plagiarism will not be tolerated. All codes should be well commented and any reference material used should be cited appropriately. If you use ChatGPT (or similar) include an explanation of how you used the tool to guide your work and/or report writing.

The in-class quiz on Assignment 2 will be on Tuesday March 3rd, 2026.

Important: Please include the following signed statement with your submission.

I, [insert name] attest that the work I am submitting is my own work and that it has not been copied/plagiarized from online or other sources. Any sourced material used for completing this work has been properly cited. [Signature]

Introduction

Variational Autoencoders (VAEs) arise from simple changes to Autoencoders. You have been provided with a Jupyter Notebook that implements the VAE found at https://keras.io/examples/generative/vae/.

Problem 1

The output of the VAE in the notebook is really just the mean of the output distribution. To produce/generate a true sample, additional steps are required. Answer the following questions:

1. What additional steps are required to use the VAE to generate a sample?
2. Augment the code to generate and display 10 random samples from a trained VAE.
3. What conclusions can you draw about the generative quality of the VAE when it is used in a "truly generative" fashion (i.e., sampling the normal distribution output of the model)? Is it common practice to use a VAE in this manner? Or does one usually just use the mean? Support your answer with references.
4. During training, what is the behaviour of the two loss terms? Does one dominate? Should it? Why or why not?
5. (Graduate Students Only) Increase the latent space from 2 to 3 and re-train the VAE. Visualize the results by showing appropriately selected slices through the latent space. Do you think there is any advantage to a larger latent space for this particular problem? Why or why not?

Problem 2

An interesting feature of AEs and VAEs is the ability to interpolate in the latent space. For example, given a VAE we might consider two random latent space locations $\underline{z}_1$ and $\underline{z}_2$. We can then vary our selection of points to decode between these two points by linearly interpolating between their positions. This idea is really interesting for, say, pictures, where we can interpolate between hair colour or between happiness and sadness. Answer the following question:

1. Demonstrate latent space interpolation for a VAE of your choice (the given VAE is sufficient). Pay special attention to how you choose your two latent space points (some approaches are more interesting than others). Provide a summary of your implementation and how it performed.

Additional Discussion Questions

1. AEs and VAEs both use a latent representation, but VAEs provide a distributional interpretation of the output. Outside of generation, are there tasks that AEs are better at than VAEs? Vice versa? Support your answer with references.

Hand-in Report

Submit a report summarizing the relevant answers to Problems 1 and 2, and the Discussion Questions. Include code snippets and explanations that facilitate our understanding of your solutions and allow us to interpret your results without resorting to running your code.

When you submit your report, also submit your code in the event that we need to run it to validate your claims.